Mansoura University
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[SWE143] Software Project Management

Lecture 02: Software Development Projects and Stakeholders

Level 300:software engineering program

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Software Development Projects and Stakeholders

The name of the game, the players, and (some of) the rules

Goals of this Unit

- Understanding what is a project, what is the life cycle of a project and how it differs from other types of works
- Understanding the players and the relationships among them
- Understanding the influences organizations exert on project and project executions

What is a project

The name of the game

A project is a temporary endeavor جهد / مسعى undertaken to create a unique product, service, or result

(definition from Project Management Institute (2004)

Characteristics of a Project

- Temporary
  - *Definitive begin* and end (either because the goals are met, or the project is closed - goals cannot or will not be met)
  - Projects’ results are not necessarily temporary (used long-term by the client)
    - follow up on defects and problems found in project outputs
- Resource constrained
  - A limited time/ resources is available to build the project outputs.
  - The output of a project is seldom the best possible solution but rather the best solution given the constraints
- Deliver output in form of
  - A <u>product</u> which is quantifiable (e.g. a software app)
  - A <u>capability</u> to perform a service, such a business function (payroll system)
  - A <u>result</u>, such as knowledge (collected in documents, presentation)

Characteristics of a Project

- Progressive elaboration التفصيل التدرجى
  - Development by steps and in increments (necessary to keep a project under scope)
  - The cost of changes increases as a project progresses
- Unique output
  - A project delivers has some novelty
  - **Risks** come from the unique characteristics of the project outputs

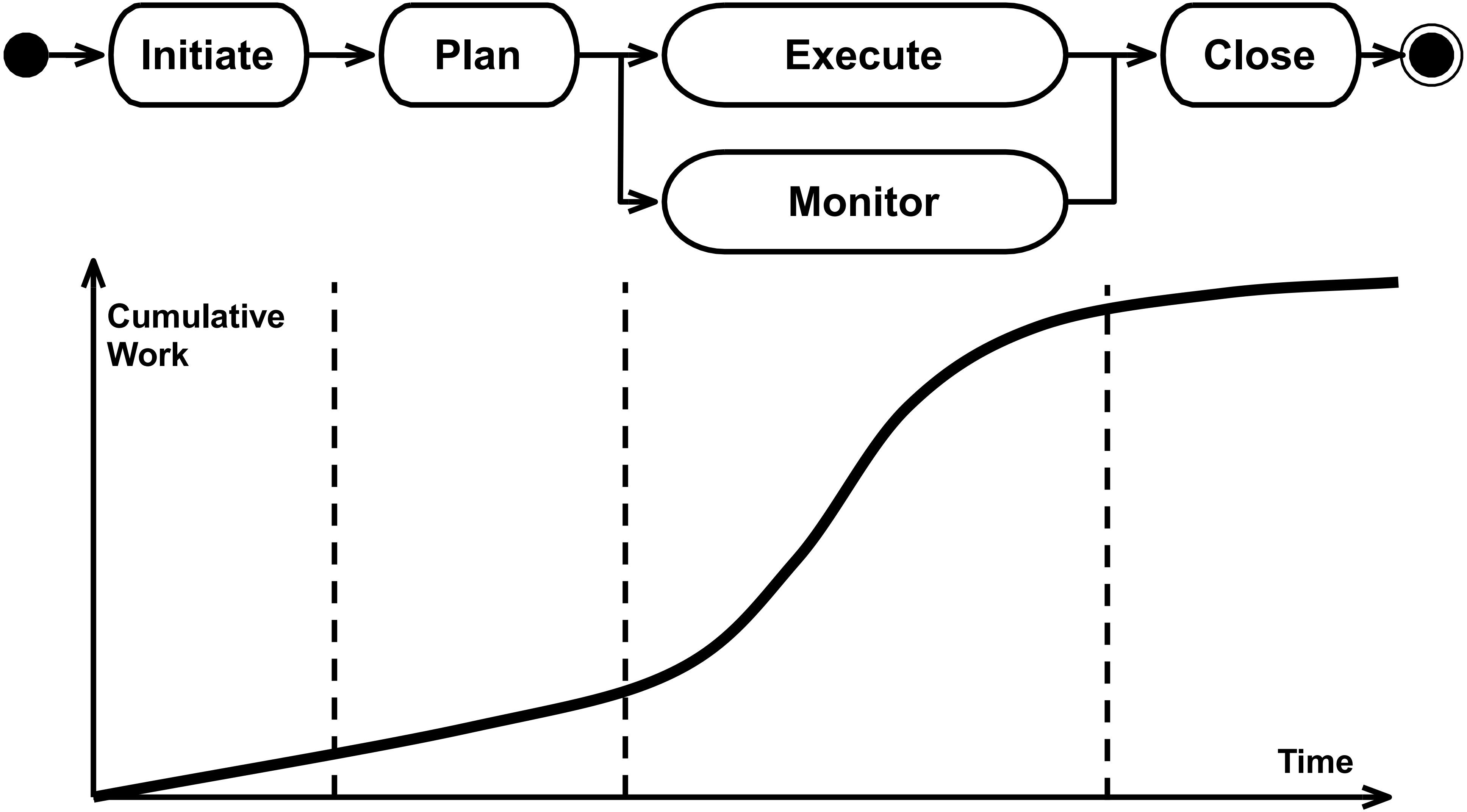
The two main types of Software Project:

- Product Development [tangible]:
  - ➢ is a type of software systems i.e., outcome of software developing process and are delivered to customer along with documentation about how to install and use particular software system.
- Service [intangible]:
  - ➢ Involves providing software as a service (SaaS) over the internet, where customers pay for access (the customer does not own the software) and the provider handles maintenance and updates.

Phases of Project (Temporary)

1. An **initiating** phase, during which the project infrastructure and the project's goals are drafted.
   - defining **what** the project is and gaining approval to move forward.
2. A **planning** phase, during which project goals are refined, activities identified and scheduled, and many other support activities are properly planned.
   - **build the roadmap** for the project. It outlines **how** the work will be done by setting clear goals, timelines, budgets, and resources..
3. An **executing** phase, during which the actual work takes place. Running in parallel, a **monitoring** phase measures the progress and raises flags when plans and reality disagree.
4. A final **closing** phase, where the project outputs are handed out and the project is closed.

Project phases and cumulative work



If we plot the cumulative work produced in a project, we get an s-shaped curve

Project Management classification

- **Program**
  - A program is a set of <u>related projects</u> manaaged in a coordinated way
  - **<u>E.g., Program consists of 3 projects</u>:** Build website / Develop mobile app / Implement cloud infrastructure
- **Subprojects**
  - Projects may be divided in subprojects (although the sub-projects may be referred to as “projects” and managed as such)
- **Portfolios**
  - A portfolio refers to a grouping of unrelated projects, and programs.
  - The purpose of a portfolio is to establish centralized management and oversight for many projects and programs.
  - **E.g., in software:** Accounting System Software Project / Mobile App Development Program / Cybersecurity Software Project

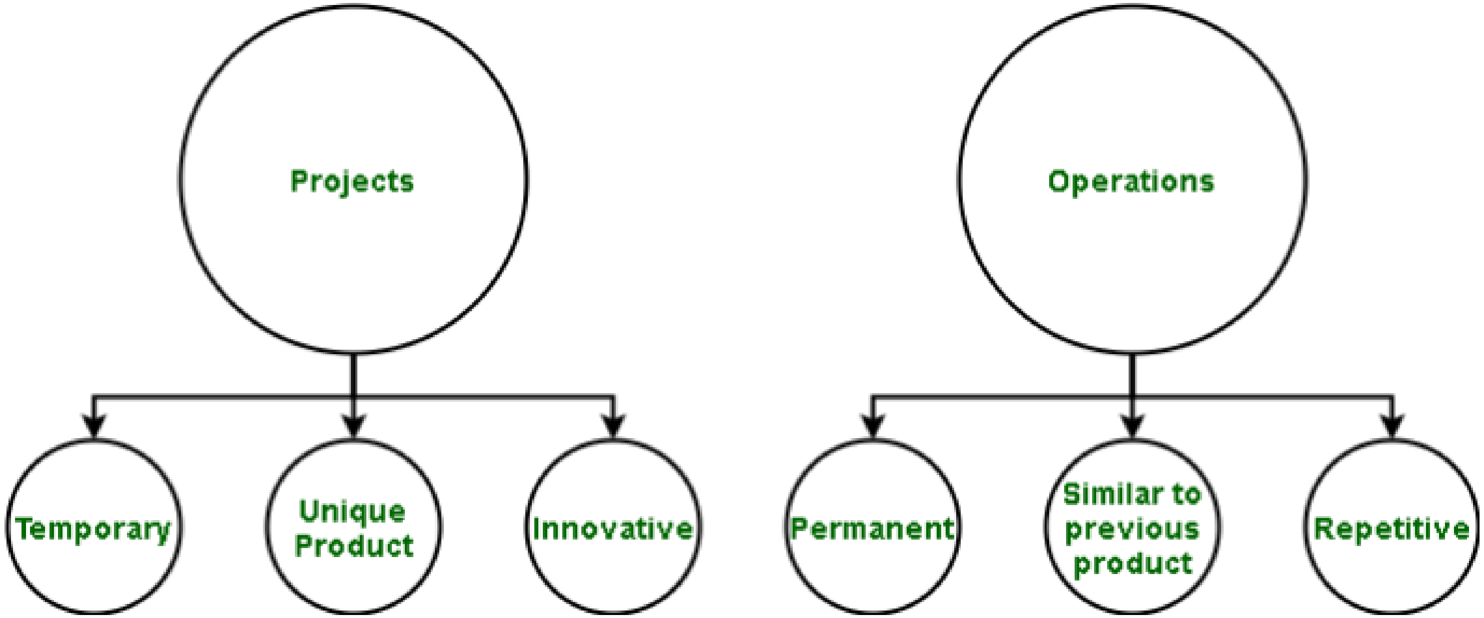
– **Project and Program Management**
  - Set of <u>related</u> projects managed in a coordinated way in order to achieve some sort of benefit
- **Portfolios and Portfolio Management**
  - Collection of <u>unrelated</u> projects or programs and other work grouped together to facilitate management and meet strategic objectives

Projects and Operational Work

- Work can be categorized either as project or operational
- Common characteristics
  - Performed by people
  - Limited resources
  - Planned, executed, and controlled
- Differences
  - Project: obtain goals and terminate
  - Operational work: sustain the business

- **For example:** The development of a new application is a project,
- and providing support to users of that application is operations.
  - **Examples of Operations** or Maintenance activities: Minor modification or an upgrade to an existing process or program, for instance

Projects and Operations



Projects and operations are both parts of the product life cycle

- **Project:** is an endeavor that is temporary in nature, that is undertaken to produce a unique product
- **Operation:** After product is made it is then brought out to market. The processes of creating product again and again using same techniques that were used earlier, for purpose of distributing to users and (making profit in most cases) are a part of Operations.

Examples (and counterexamples)

- Cooking dinner
  - **Type** is project: Cooking dinner is a one-time task with a clear start and end.
  - **Counterexample** of operations: Running a restaurant's daily meal service is ongoing and repetitive.
- Building a car
- Designing a car
- Writing a paper
- Developing a software system

○ Maintaining a software system

○ Managing personnel

Software Development Projects

Some Examples of Software Development Projects and Operational Work

Type of "Software" Development Projects

- In your life as a project manager you might be involved in different types of "software" development projects, among which:
    1. Application Development
    2. Process and Systems Re-Engineering
    3. System Integration
    4. Consulting Services
    5. Installation and Training

Application Development

- Goal: developing an application (desktop, web, mobile, embedded)
- Types of application development:
  - **One-offs:** (**bespoke** systems) systems specifically created for a client
    - Examples: a luggage tracking software, a system to monitor a fleet of trucks.
    - The ownership and the source code of the final product might also be handed over to the customer
    - Offers an opportunity for the supplier to enter a new market or establish a long-term relationship with a new customer.
    - The novelty of the product also constitute the main risk both for the customer and the supplier.

1. Application Development

- **Off-the-shelf:** to fill the need of a large set of users
  - Origin: the company developing the system, which sometimes conducts user surveys, to better understand needs and features that are most useful.
  - The company developing the system sets the roadmap, choosing when to upgrade, what functions to add, and when to do maintenance.
- **Customized off-the-shelf:** standardized systems which require a significant amount of customization to be used in an organization.
  - Example: **Enterprise Resource Planning** (ERP) systems

How do ERP systems work?

An ERP (Enterprise Resource Planning) system – also called an ERP suite – is made up of integrated modules or business applications that talk to each other and share common a database.

Each ERP module typically focuses on one business area, but they work together using the same data to meet the company's needs. Finance, accounting, human resources, sales, procurement, logistics, and supply chain are popular starting points. Companies can pick and choose the module they want and can add on and scale as needed.

2. Process and Systems Re-Engineering

- Goal: change the way in which the operational work of an organization is carried out to achieve some strategic goal (e.g., improve quality, become more efficient)
- In many cases, the system being introduced is an ERP
- Typically, large projects which involve an accurate analysis of the existing situation (“as is”) w.r.t. procedures, systems, infrastructure
  - Based on the analysis, the project identifies **interventions** that will improve processes, reduce bottlenecks, or automate tasks, and implement those changes.

3. System Integration Services

- Goal: automating the information flow among the systems of an organization. Improve the efficiency of work and to reduce data duplication and errors
- The approach is chosen when migrating to a new system is impractical or too costly
  - Instead of completely replacing legacy systems, this approach integrates them into a more efficient, cohesive framework.

- Types of integration:
  - **Vertical :** integration of systems performing similar operations [ensures all departments have access to **consistent, up-to-date customer data**, avoiding duplicate entries]
    - E.g., putting together data about customers kept by different departments.
  - **Horizontal :** integration of systems automating different steps of a procedure [reducing delays and errors caused by manual handoffs between departments]
    - E.g., automating the flow of orders from marketing to production.

Other types of Projects

4. Consulting Services
   - Typically asked to gain a know-how outsize a company's core competence اكتساب معرفة تتجاوز قدرة الشركة
     - Consulting services are expert services provided to organizations when they need specialized knowledge or skills that are not part of their core business.
   - E.g., Evaluating the reliability of a software system using advanced techniques that a company's internal team may not specialize in.
5. Installation and Training Services
   - Services related to the installation or training on specific software systems
   - A revenue model in open-source development
     - **The open-core model** is a business model for the monetization of commercially produced open-source software. It involves offering a "core" or feature-limited version of a software product as free and open-source software, while offering "commercial" versions or add-ons as proprietary software

Projects and their Environment

The players (and you)


Project
Stakeholders

A **project stakeholder** is any individual or an organization that is actively involved in a project, or whose interest might be affected (positively or negatively) as a result of project execution or completion.

(definition from Project Management Institute (2004)

The Players

- Some characteristics:
  - They may have different influence and varying level of responsibility during the project
  - They may play different roles
  - They may have positive or negative influence on the project
  - They may be difficult to identify
  - Their lack of intervention may negatively influence the project (need for identification and involvement)
- Remark: the project manager and the project team **are** project stakeholders, although the term is often used to refer to the "other" stakeholders (e.g., customers, sponsors)

Stakeholder Management Process for Project Manager

Identify stakeholders: Find all stakeholders (internal and external).

Analyze and prioritize: Assess their influence and interest.

Identifying Risks and Opportunities: define how to engage each stakeholder/Communicate regularly and address concerns proactively.

Relationship Management: Track engagement and adjust strategies when necessary.

Types of Stakeholders

- The project manager
  - has the responsibility of project planning, procuring الحصول على, and executing the project.
- The project team
  - responsible for carrying out the work in a project
- The project sponsor:
  - responsible for approving, funding, and ensuring the success of a project, even without direct management authority
- The performing organization:
  - is the primary group that is most directly involved in the project, and is responsible for enabling success. Therefore, it can be considered as a superset of some of the most crucial stakeholders
- The client: benefits from the project outputs
- The “rest”: anyone who might be affected by the project outputs


STAKEHOLDER
A person or group with an interest in something, such as a business or institution.
INTERNAL
An internal stakeholder is anyone who has a direct interest in you or your organization. The stakeholder will be directly affected by the success or failure of the organization. Common examples include employees, shareholders, and owners.
EXTERNAL
An external stakeholder is a person or organization who has an interest in the success or failure of a project, business, or organization but is not directly involved in its operations. Common examples include customers, governments, local business groups, and trade unions.

Key Stakeholders

- Internal:
  - Project team members: the group performing the work
  - Project management team: the members of the team directly involved in project management
  - Sponsor: person or group providing the financial resources

- External:
  - Customer/User: person or organization that will use the results of a project.
  - Influencers: people or groups not directly related to the project who could influence the course of a project

Stakeholder Identification Exercise

- Identify the stakeholders in Education

  - **Students**
  - **Parents**
  - **Teachers**
  - **School Administrators**
  - **Business Community**
  - **Taxpayers**
  - **Unions**
  - **Governments**
  - **Universities**

Organizing the Development of Software Projects

Software Project Management

- **Software project management** is the integration of management techniques to software development.
- The need for such integration has its root in the sixties, in the days of the "**software crisis**", when practitioners recognized the increasing complexity of delivering software products meeting the specifications
  - The crisis highlighted that many software projects failed to meet expectations due to issues like scope creep (project's requirements keep increasing or changing after the project has already started) توسع, missed deadlines, and poor communication.

What makes a Software Product? Examples:

- An application to keep a list of movies watched
- A double-entry accounting system for home or a small businesses
- A web service to manage the reservations of squash courts in a squash club
- A system to plan the resources of a big corporation
- An automated breaking system for a train

Think about software products in terms of:

- What components does it have? (UI, API, database, etc.)
- What happens if it fails? (impact on users or business)
- What skills are needed to build it?
- What business model supports it?

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A software product is not just a program; it is a complete
system made of many parts, designed for real-world use to
solve problems, and shaped by both technical and business
considerations.
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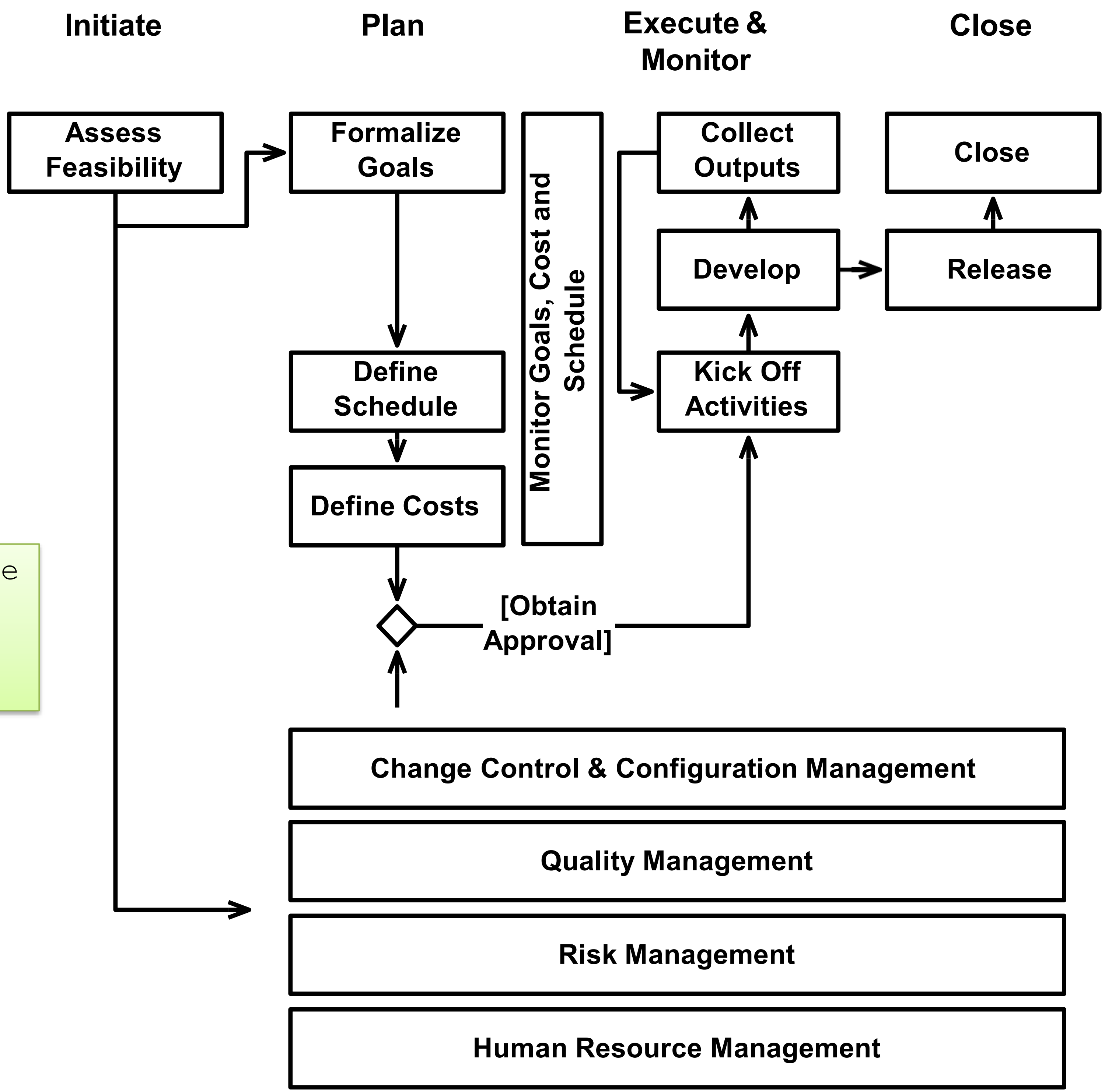
Software Development Framework

- A general software project management framework is meant to:
  - Form a shared vision about the goals to be achieved, the characteristics of the project outputs, and the characteristics of the development process
  - Structure the work as a progressive refinement, from specification to goals
  - Reduce the impact of uncertainties and unknowns
  - Highlight any deviation from the plan (goals, costs, quality)
  - Ensure the coherency and quality of the project artifacts over time and in spite of unknowns and (request for) changes
  - Motivate your team

Some Concerns

- Feasibility Assessment
- Goals (Scope) Management
- Time Management
- Cost Management
- Change Control and Configuration Management
- Quality Management
- Risk Management
- Human Resource Management



The software project management Layers

Thank You